MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10547250 (MAIN - A5)

1. SCOPE

1.1 This specification covers a multi-layer circuit board assembly, a subassembly of the Counter, Time Interval: 10547200 which is a modular component of the Range Finder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical, for Coating Printed Circuit Assemblies
MIL-P-46843	Printed Circuit Assemblies, Design and Production of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWING

U.S. Army, Frankford Arsenal

F10547250	Circuit Card Assembly - Main - A5

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PACKAGING DATA SHEETS

10547250	Packaging of Circuit Card Assembly - Main - A5

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The Circuit Card Assembly - Main - A5, hereinafter referred to as assembly, shall be manufactured in accordance with Drawing F10547250 and drawings pertaining thereto.

3.3 General specification.- The assembly shall meet the following requirements, where applicable, of MIL-F-13926 and MIL-P-46843:

a. Dimensions and tolerances
b. Inorganic protection surface finisher
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meeting the workmanship requirements of MIL-F-13926 and MIL-P-46843, the assembly shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.11.4 inclusive.

3.3.1.1 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with at least one point of the component in contact with the board. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface. Components stacked directly above another component shall be mounted with at least one point of the upper component in contact with the bottom component. The upper component shall not be raised more than 0.031 inch from the bottom component at any point on its contacting surface.

3.3.1.2 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that the proper identification markings on each part are visible.

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3.3.1.3 Bend radius.- The minimum bend radii for formed leads shall be one lead diameter. The maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.1.4 Component leads.- The minimum lead length for radial lead components shall be 4 times the lead diameter. The portion of the lead between a bend and the component body shall emerge from the body in a straight line radially from the component body and enter the plated hole approximately perpendicular to the board. Component leads shall be preformed to suit mounting holes and clinched parallel to board after being inserted through the holes. There shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than .060 inch, including solder, beyond the side of the circuit board opposite the component. The leads shall not extend more than two lead diameters beyond the land area.

3.3.1.5 Soldering.- All solder joints shall conform to MIL-F-13926 and MIL-P-46843. There shall be a solder fillet between the board and the lead on the side of the board opposite the component. The fillet shall not obscure the contour of the clinched lead. A solder fillet is acceptable but not required, on the component side of the board.

3.3.1.6 Plated-through connections.- All plated-through connections shall be filled with solder in order to eliminate dependence on plating itself for circuit continuity.

3.3.1.7 Electrical clearance.- A minimum of .010 inch shall be maintained between uncommon component leads.

3.3.1.8 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed. The cleaning fluid shall have a gentle flow at no time exceeding a pressure of 5 pounds per square inch (psi). The temperature of the cleaning solution shall not exceed 125 degrees Fahrenheit (°F). After cleaning, there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaminated after cleaning, nor shall markings become illegible.

3.3.1.9 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than the room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements:

a. Permissible dust particle size not to exceed 6 microns.
b. Permissible moisture content limited to 2.12 grams per cubic foot (dew point 32°F maximum).
c. Dust count shall not exceed 1000 particles per cubic foot.

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3.3.1.10 Coating.

3.3.1.10.1 Masking.- Prior to coating, the areas that are to be free of coating shall be masked as specified on Drawing F10547250. After coating, all specified areas shall be clean and free of coating.

3.3.1.10.2 Coating.- The assembly with components installed shall be coated with insulating compound, electrical, Type PUR Polyurethane Resin conforming to MIL-I-46058.

3.3.1.10.3 Thickness.- Multiple applications shall be used with proper curing between each coat until a minimum of .001 inches thick to .006 maximum thickness is achieved.

3.3.1.10.4 Cured compound.- The cured compound shall present a smooth continuous surface and shall pass from the mounted components to the circuit board with no break at their junctions. The cured compound shall meet the following requirements:

a. Compound shall be free of visible dust particles or other foreign matter.
b. It shall be smooth and uniform and have no globules or bare spots.
c. It shall be substantially free of bubbles.
d. There shall be no blisters or cracks.
e. It shall be transparent and shall not affect marking legibility.
f. It shall be capable of being removed in local areas for replacement of components by means of a soldering iron or similar method.
g. Applied to a test board of 1/16 inch thickness, the compound shall not rupture or separate from the board, when the board is bent into an arc of twelve inches radius.

3.3.1.11 Wire termination.- When soldering wires to terminals or solder points, a small gap of 1/32 inch minimum to 1/16 inch maximum shall be left between the wire insulation and the solder build-up. Burned, frayed, or otherwise damaged wire insulation shall not be acceptable. Stripping of insulation shall be accomplished without disarranging, cutting, or otherwise damaging any strand of the conductor. Heat shall not be used to remove the outer jacket from shielded wire. Only mechanical methods of removal shall be permitted.

Exception: The hot wire stripping method may be used where the wire or cable primary insulation and jacket are made of Teflon (TFE) or equal.

3.4 Vibration.- The assembly shall withstand vibrations for 15 minutes in each of 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500

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cycles per second (cps). The double amplitude of the applied vibrations shall be .036 inch at lower frequencies and shall be maintained at, but not exceed, a 5 G level at higher frequencies. Subsequent to vibration, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.5 Shock.- The assembly shall withstand a total of 6 shock impulses, one in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 millisecond. The peak amplitude of each shock impulse shall be 75 G's. Subsequent to shock, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.6 Storage temperature.- The assembly shall withstand thermal stabilization at temperatures of -80°F and +160°F. Subsequent to exposure to the temperatures specified herein, the assembly shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.8.

3.7 Performance.

3.7.1 Operating temperature.- The assembly shall meet the requirements of 3.7.2 and 3.8 when thermally stabilized at all temperatures from -65°F to +155°F inclusive.

3.7.2 Operation.- With +160 ± 5VDC applied between E1 (+) and E18 (ground), the numerical indicator or neon lamp of Column B of Table I shall glow when the corresponding connector pin of Column A of Table I is grounded.

TABLE I

"A" Connector - Pin	"B" Numerical indicator or neon lamp display
J1 - Q	V1 - 0
G	1
O	2
E	3
M	4
C	5
A	6
N	7
B	8
P	9
J2 - n	V2 - 0
Y	1
o	2
X	3
m	4
V	5
k	6
T	7
i	8

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TABLE I Continued

"A" Connector - Pin	"B" Numerical indicator or neon lamp display
J2 - S	V2 - 9
J3 - b	V3 - 0
a	1
S	2
W	3
R	4
V	5
Q	6
Z	7
P	8
T	9
J4 - Z	V4 - 0
a	1
h	5
U	7
J3 - U	2
J4 - I, J4 - d, J4 - S	DS 1
J4 - H, J4 - I, J4 - S	DS 2
J4 - F, J4 - H, J4 - I	DS 3
J4 - J, J4 - F, J4 - H	DS 4
J4 - d, J4 - J, J4 - F	DS 5
J4 - J, J4 - d, J4 - S	No neon bulb shall glos

3.8 Electrical circuitry.- The electrical circuits of the assembly shall conform to the electrical schematic drawing and the requirements of Table II.

TABLE II

Connector	Pins		Terminal Point		Resistance or Capacitance (Nominal)
	From	To	From	To	
J4	I	H			122K ohms (1)
J4	H	F			122K ohms (1)
J4	F	J			122K ohms (1)
J4	J	d			122K ohms (1)
J4	d	S			122K ohms (1)
			E8	E30	1K ohms
			E10	E9	4.7K ohms
			E7	E13	1K ohms
			E7	E12	2.2 microfarad
			E6	E10	2.2 microfarad
			E5	E33	.02 microfarad

(1) Diode in series, reverse meter leads if necessary.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III, IV, V and VI herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables III, IV, V and VI shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE III - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% DEFECTIVE		
101. Spacing of parts	3.3.1.1	SME; Visual
102. Symbols and markings	3.3.1.2	Visual
103. Bend radius	3.3.1.3	SME; Visual
104. Component leads	3.3.1.4	SME; Visual
105. Soldering	3.3.1.5	Visual
106. Plated-through connections	3.3.1.6	SME; Visual
107. Electrical clearance	3.3.1.7	Visual
108. Masking	3.3.1.10.1	Visual
109. Cured compound	3.3.1.10.4	SME; Visual
110. Wire termination	3.3.1.11	Visual
MINOR: NONE		

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

TABLE IV - CLASSIFICATION OF DEFECTS
(100% Inspection)

	REQUIREMENT	TEST PROCEDURE
111. Operation	3.7.2	4.6.4.2
112. Electrical circuitry	3.8	4.7

NOTE: The tests in Table IV shall be conducted at a temperature between +70°F and +80°F.

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4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.9
Drying	3.3.1.10
Coating, insulation compound	3.3.1.11

4.4 Special sampling.

4.4.1 General.- Three assemblies shall be selected at random as a special sampling from each 50 produced. The samples shall meet the requirements and tests in Table V.

TABLE V

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable Dwgs & Specs - Visual
302.	Fabrication	3.2	Applicable Dwgs & Specs - Visual
303.	General specifications	3.3	MIL-F-13926, MIL-P-46843 Visual

NOTE: The tests in Table V shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- Three assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table VI and shall then meet the requirements and tests of Table IV.

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TABLE VI

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration (+70°F and +80°F)	3.4	4.6.3
305.	Shock (+70°F and +80°F)	3.5	4.6.2
306.	Storage temperature (-80°F and +160°F)	3.6	4.6.1.1
307.	Operating temperature (-65°F and +155°F)	3.7.1	4.6.4.1, 4.6.4.2

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

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4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests specified in 4.6.

Nomenclature	Description
1. D. C. Power supply	160 ± 5 VDC, current rating 10 ma

4.6 Test methods and procedures.

4.6.1 Environmental.- The assemblies shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the assemblies shall be subjected to the tests specified in 4.6.4 through 4.7.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.6.

4.6.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.5.

4.6.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The amplifier shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.4. The vibration test shall then be performed in accordance with Method 514, Procedure 1, Part 1 of MIL-STD-810.

4.6.4 Performance.- Connect the positive side of the +160 VDC power supply (Item 1 in 4.5.3) to E1 and the power ground to E18. The above connections shall remain for all subsequent tests for compliance with 3.7.2.

4.6.4.1 Operating temperatures.- The assembly shall be thermally stabilized at the temperature extremes specified in 3.7.1 utilizing the procedures specified in 4.6.1.1.The assemblies shall meet the requirements specified in 3.7.2.

4.6.4.2 Operation.- With the power supply (Item 1 in 4.5.3) connected as specified in 4.6.4, ground the pin (or pins for neon lamps) as specified in Column "A" in Table I of 3.7.2 and observe that the numerical

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indicator tube grid or neon bulb glows as specified in Column "B". The assemblies shall meet the requirements specified in 3.7.2.

4.7 Electrical circuitry.- Continuity shall be inspected with standard measuring equipment (SME) for compliance with Drawings D10547257 and F10547250 and Table II as specified in 3.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10547250. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Circuit Card Assembly, Main - A5, provides a numerical readout of target range and indicates the number of targets received. It is a sub-assembly of the Counter, Time Interval which is a module of the Range Finder, Laser, AN/GVS-3.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable packaging data sheet.
c. Selection of applicable levels of preservation, packaging and packing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A201

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